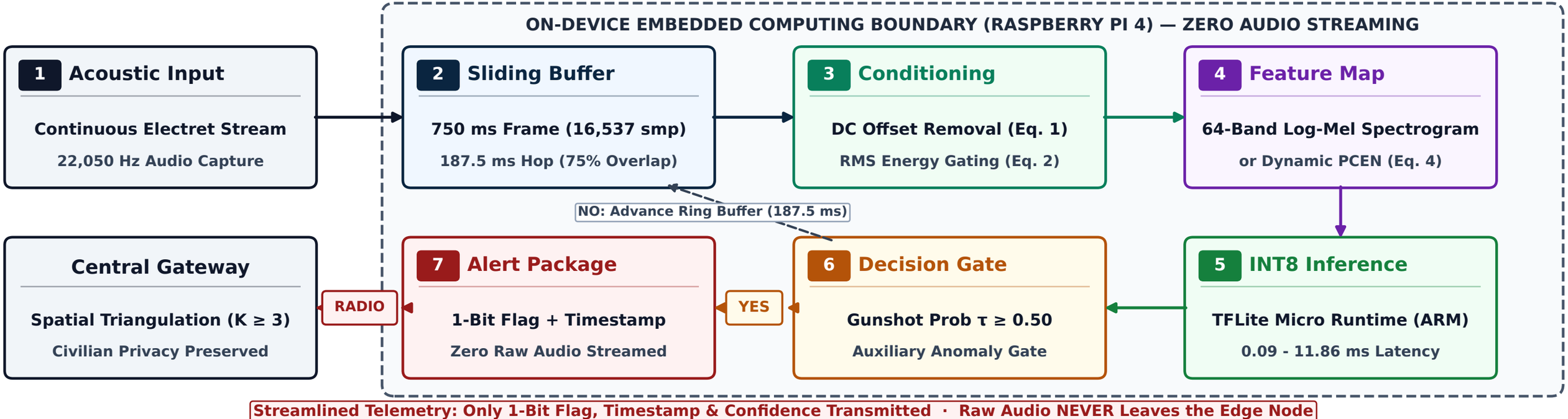
ON-DEVICE EMBEDDED COMPUTING BOUNDARY (RASPBERRY PI 4) — ZERO AUDIO STREAMING
1 Acoustic Input
Continuous Electret Stream
22,050 Hz Audio Capture
2 Sliding Buffer
750 ms Frame (16,537 smp)
187.5 ms Hop (75% Overlap)
3 Conditioning
DC Offset Removal (Eq. 1)
RMS Energy Gating (Eq. 2)
4 Feature Map
64-Band Log-Mel Spectrogram
or Dynamic PCEN (Eq. 4)
5 INT8 Inference
TFLite Micro Runtime (ARM)
0.09 - 11.86 ms Latency
6 Decision Gate
Gunshot Prob τ ≥ 0.50
Auxiliary Anomaly Gate
NO: Advance Ring Buffer (187.5 ms)
YES
7 Alert Package
1-Bit Flag + Timestamp
Zero Raw Audio Streamed
RADIO
Central Gateway
Spatial Triangulation (K ≥ 3)
Civilian Privacy Preserved
Streamlined Telemetry: Only 1-Bit Flag, Timestamp & Confidence Transmitted · Raw Audio NEVER Leaves the Edge Node